Comparison Between Scientific Reports (2025) Paper and Your Project

1. Core Focus Comparison

Aspect	Scientific Reports Paper (2025)	Your Project
Detection Type	Phishing website detection	Phishing + Spam + Legitimate classification
Input Type	URL-based features only (30 handcrafted features)	URL + Email Content + Header features
Dataset Size	11,055 URLs (UCI dataset)	Multi-source datasets (potentially 30k–200k samples)
Classification	Binary (Phishing / Legitimate)	Multi-class (Phishing / Spam / Legitimate)
Best Model	Random Forest + SHAP	Ensemble ML (RF / XGBoost) + SHAP + LLM

2. Explainability Depth

Their Paper:
- Uses SHAP for feature importance ranking
- Focuses on identifying top 15 URL features
- Stops at technical interpretation (SHAP plots)

Your Project:
- Uses SHAP for mathematical explanation
- Uses LLM to convert SHAP output into human-readable explanation
- Adds trust score
- Adds risk score
- Adds phishing category classification
- Adds user feedback loop

3. Architectural Difference

Their Framework:
Data → Preprocessing → ML model → SHAP → Performance metrics

Your Framework:
Data → Feature Extraction (URL + Content + Header)
→ Ensemble ML
→ SHAP
→ LLM Explanation
→ Risk Score

→ Trust Score
→ Phishing Category
→ User Feedback Loop

4. Performance & Scope

Their Paper:
- Accuracy: 97% (RF + SHAP)
- Binary detection
- Focus on computational efficiency
- Lightweight deployment emphasis

Your Project:
- Multi-class classification (harder task)
- Risk-aware output
- Trust-aware scoring
- Human-interpretable explanation
- Broader data scope (email + header + URL)

5. Research Gap Between Them & You

Their limitations:
- Only URL-based
- No multi-modal data
- No spam differentiation
- No trust modeling
- No human-centered explanation
- SHAP computational cost noted

Your project addresses:
- Multi-source features
- Multi-class detection
- LLM-driven interpretability
- Trust-aware scoring
- Risk quantification
- User feedback loop

6. Clear Summary of Differences

Feature	Their Paper	Your Project
Binary classification	Yes	No

Feature	Their Paper	Your Project
Multi-class (Spam included)	No	Yes
URL-only features	Yes	No
Email + Header features	No	Yes
SHAP explainability	Yes	Yes
LLM explanation	No	Yes
Risk scoring	No	Yes
Trust scoring	No	Yes
Phishing category classification	No	Yes
User feedback loop	No	Yes
Decision-support orientation	Partial	Strong